ATTILI KIREETI ADITYA KUMAR

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EDUCATION

B.Eng (Hons), Electrical & Electronic Engineering

Nanyang Technological University • Singapore 08/2023 - 06/2027

- Specialisation: Data Analytics & ML
- Key Modules: Data Science & AI, Data Structures & Algorithms (Python), Digital Signal Processing, Signals & Systems, Language & the Computer (NLP), Venturing into Entrepreneurship

SKILLS

Programming & Tools: Python, SQL, GitHub, Power BI, Vercel, C#, Unity

Data Analysis Tools: Pandas, NumPy, Matplotlib, Scikit-learn, Excel, Seaborn, NLTK, CoNLL-U, TF-IDF

ML & AI: LLM API Integration (OpenAI, Claude, Groq), NLP Pipelines, OCR, Prompt Engineering, Data Preprocessing, Regression & Classification Models

Hardware: Arduino, ESP32, Raspberry Pi, Embedded Systems

WORK EXPERIENCE

SMRT Corporation • Singapore 01/2026 - Present

Executive Engineer • Internship

- Built an ETL analytics pipeline for Automatic Train Supervision (ATS) server logs, parsing multi-source telemetry into a unified, lean dataset for real-time health monitoring, anomaly detection, and root-cause analysis via Power BI dashboards.
- Developed PowerShell scripts to analyze Wayside Radio Units' health, enabling remote fault detection and reducing maintenance delays within 3 months.

PROJECTS

Rule-Based POS Tagging Pipeline (Pengo) 09/2025 - 10/2025

HG2051 Language & the Computer | Individual Project

- Engineered a deterministic NLP pipeline for POS-tagging a Pengo CoNLL-U corpus: rule engine (JSON config + regex), gold-standard overrides, and post-processing validation.
- Built corpus comparison tooling to align before/after annotations and generate automated UPOS accuracy reports.

JAGA – Smart Fall Detection for Elderly 06/2025 - 07/2025

Huawei Tech4City Hackathon 2025 (Smart Healthcare)

- Prototyped privacy-first fall detection using ESP32-S3 with AMG8833 thermal array and INMP441 mic; designed edge-first architecture with local inference and community alarm integration.

PatchPal – AI-Driven Code Optimisation Pipeline 03/2025

iNTUition Hackathon 2025

- Built end-to-end pipeline: ingested user feedback into SQL, applied NLP text cleaning, vectorised with TF-IDF, and clustered issues using scikit-learn for automated prioritisation.
- Integrated LLM API to generate targeted code fixes from clustered feedback, then auto-compiled and validated patches before surfacing only passing fixes for developer approval.

AWARDS & SCHOLARSHIPS

Ideasjam 2026, IHH Healthcare Sponsor Award – ALLY (AI Caregiver Co-Pilot) 01/2026
SUNIG 2025, 2nd Runner-up Varsity Tchoukball 09/2025
Temasek Foundation TF LEaRN Scholarship 2025, ASEAN Summer Program NTU 06/2025
iNTUition Hackathon 2025, 2nd Runner-up (PatchPal) 03/2025
Escendo Make-a-thon 2025, 2nd Runner-up (Shelfie) 01/2025
SUNIG 2024, 1st Runner-up Varsity Tchoukball 09/2024

VOLUNTEERING & LEADERSHIP

Nanyang Technological University
10/2024 - 10/2025

Vice-President of Programmes (Efinity Orientation, EEE)

- Led planning and execution of a 3-day orientation for 388 freshmen with 160 committee members
- Supervised 6 Chief Programmers, 2 Logistics Directors, and a Safety Director to ensure smooth operations
- Conducted interviews and selected next-gen leaders; planned a senior camp for 163 members

AIESEC Global Classroom Programme
12/2024 - 01/2025

Global Volunteer Teacher • Heping, Taichung, Taiwan

- Taught 40+ junior high students in Math and English, achieving 10–20 % academic improvement.
- Designed and executed lesson plans across multiple classes; promoted cross-cultural learning.
- Strengthened communication and adaptability while teaching in a multilingual environment.

NTU EEE Overseas Community Immersion Programme
02/2024 - 07/2024

Programmes and Cultural Exchange Sub-Committee Member • Ho Chi Minh, Vietnam

- Organized bonding activities for 27 volunteers and created lesson plans for 20–30 students.
- Coordinated a multicultural fashion show for 150 guests to showcase Singaporean culture.
- Deepened appreciation of Vietnamese culture through 13 days of local collaboration.